MOBINA SAEDI

Computer Engineering Undergraduate | Aspiring AI/ML Researcher

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EDUCATION

B.Sc. in Computer Engineering – Islamic Azad University, Central Tehran Branch Expected Graduation: June 2027

GPA: 18/20

Selected coursework (grades out of 20)

Course	Grade	What it covered / what I did
Software Project	**20/20**	Defended my research manuscript as the project: a paired protocol testing how Persian orthographic variants (Arabic-Persian yeh and kaf) and input normalization affect LLM multiple-choice question answering, with a Python evaluation pipeline and a pilot experiment. The manuscript is in preparation.
Internship Project	**20/20**	Built Air Drawing, a real-time computer-vision application (Python, OpenCV, MediaPipe) that tracks the hand through a webcam and turns finger gestures into drawing on a virtual canvas.
Theory of Languages and Compilers	**20/20**	Formal languages and automata, lexical analysis, parsing, and the phases of compiler construction.
Algorithms	**17/20**	Design and analysis of algorithms: complexity, divide and conquer, greedy methods, dynamic programming, and graph algorithms.

RESEARCH INTERESTS

Machine Learning, Artificial Intelligence, Deep Learning, Natural Language Processing (NLP), Large Language Models (LLMs), Explainable AI (XAI), Low-Resource and Multilingual NLP, Data Science

RESEARCH PUBLICATIONS

Orthographic Variation and Input Normalization in Persian LLM Multiple-Choice Question Answering: A Paired Pilot Protocol Manuscript in preparation – 2026

Mobina Saedi*, Mohammad Farshad Fazeli Sabzevar* (*co-corresponding authors). Full text: mobinasaedi.github.io/Orthographic_Variation_Persian_LLM_MCQA.pdf

- Asks each multiple-choice question under six conditions (as written, with a controlled Persian spelling change, and after each of two normalizers) to test whether Arabic and Persian forms of yeh and kaf, or ZWNJ versus space, change LLM answers.
- Pilot with Qwen2.5-1.5B-Instruct (llama.cpp): 2,982 requests on 97 yeh/kaf and 69 ZWNJ items. Aggregate accuracy was identical before and after the change, while individual answers flipped; the paper states its limits (one small quantized model, author-approved items).
- My contribution: writing the manuscript, designing the paired protocol, and implementing the Python code.

Status: in preparation; not yet published or submitted.

PROJECTS

Air Drawing – Gesture-Based Virtual Drawing System (code) Python, OpenCV, MediaPipe

- Real-time touchless drawing: MediaPipe hand tracking turns finger gestures (draw, erase, change color, clear, pause) into actions on a virtual canvas.

License Plate Detection (code) Python, OpenCV

- Classical computer-vision baseline that localizes a possible license plate region: grayscale, noise reduction, Canny edges, contour approximation, and filtering by aspect ratio and area. It does not read characters.

Cat Detection and Counting (code) Python, YOLO, OpenCV

- Detects cats with a pretrained YOLO model, counts the detections, and saves an annotated image, with a configurable confidence threshold.

EXPERIENCE

AI/ML Intern – AIO Learn Oct 2026 – Present

- Building foundational machine learning knowledge through structured learning and practical training.

English Language Instructor and Online English Tutor (including freelance tutoring) Jun 2024 – Present

Safir Gofteman Institute, Melal Language Institute, AmazingTalker

- Teaching English to learners of different ages and proficiency levels, in institutes and online.

TECHNICAL SKILLS

Programming: Python
Data analysis: NumPy, Pandas
Visualization: Matplotlib, Seaborn
Machine learning: Scikit-learn; foundational knowledge of machine learning concepts
Computer vision: OpenCV, MediaPipe, YOLO (Ultralytics)
Tools: Jupyter Notebook, Git/GitHub, llama.cpp (local LLM inference)

CERTIFICATIONS

Exploratory Data Analysis for Machine Learning – IBM, Coursera Issued August 2026
Verify: coursera.org/verify/D792BPCQ0W14

Programming with Python – Mojtama Fanni (Technical and Vocational Training Complex) Issued 2024

IBM Machine Learning Professional Certificate In progress

LANGUAGES

Persian: Native
English: Advanced (C1, self-assessed). IELTS preparation in progress; target test date Spring 2027.
French: Upper-intermediate (B2, self-assessed). TCF preparation in progress; target test date Spring 2027.

I have teaching and freelance experience as an English instructor. I have not yet taken IELTS because sanctions on Iran restrict access to international testing; I am now preparing for IELTS and for the TCF.